

# SRv6 L3VPN, shown on real boxes

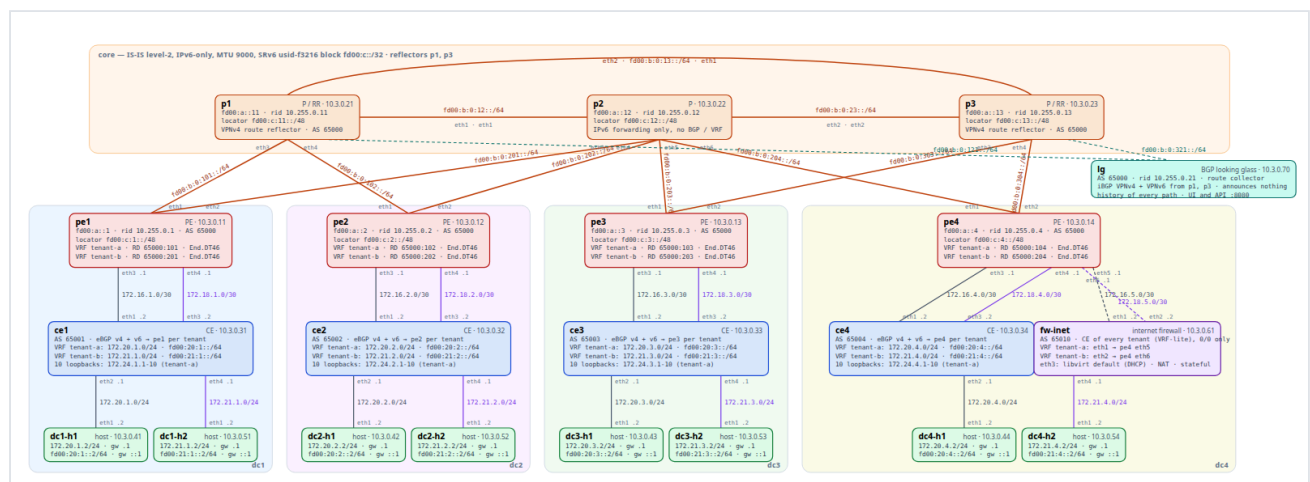
A walkthrough of the `srv6-core lab`: how an IPv6 segment-routing core carries dual-stack BGP VPNs, traced packet by packet on VyOS routers you can boot yourself. Every command output below was captured from the running lab ( `docs/walkthrough_capture.py` ); nothing is retyped.

## 1. What SRv6 is, in one paragraph

Segment Routing over IPv6 puts the forwarding instructions **inside the IPv6 destination address**. A router advertises a prefix — its *locator* — and every address under it is a *SID*: "an instruction I will execute if a packet arrives addressed to me". `End` means "forward me on", `End.X` means "send me out this link", `End.DT46` means "strip the IPv6 header and look the inner packet — IPv4 or IPv6 — up in this VRF". A packet that must visit several routers carries the list in a *Segment Routing Header* (SRH); a packet that needs only one instruction carries none — the SID is the destination. There is no MPLS, no LDP, no RSVP, no tunnel state in the core: the P routers just route IPv6.

With the **uSID** (micro-SID) flavour used here, one 128-bit address packs several 16-bit instructions, so even a multi-hop explicit path fits in the destination address alone. The lab shows both forms.

## 2. The lab



| Layer    | What                                                                                          | How                                                  |
|----------|-----------------------------------------------------------------------------------------------|------------------------------------------------------|
| Core     | 3 P routers in a triangle (p1, p2, p3), 4 PEs, one per data centre; IPv6-only links, MTU 9000 | VyOS rolling (FRR 10.6, Linux 6.18)                  |
| Underlay | IS-IS level-2 carrying loopbacks <b>and SRv6 locators</b>                                     | protocols isis ... segment-routing srv6 locator main |

| Layer      | What                                                                                                                                            | How                                                                          |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| SRv6       | block <code>fd00:c::/32</code> , /48 locator per node, uSID <code>usid-f3216</code> (32-bit block, 16-bit node, 16-bit function)                | <code>protocols segment-routing srv6 locator main</code>                     |
| Overlay    | BGP VPNv4 <b>and</b> <b>VPNv6</b> between PE loopbacks, route reflectors on p1 <b>and</b> p3, SIDs carried in the BGP Prefix-SID attribute      | <code>address-family ipv4-vpn / ipv6-vpn, sid vpn per-vrf export auto</code> |
| Tenants    | <code>tenant-a</code> (VRF table 100, RT 65000:100) and <code>tenant-b</code> (200, 65000:200), each with one CE-attached dual-stack LAN per DC | VRF per tenant on PEs and CEs; one eBGP session per address family           |
| Sites      | CE per DC (VyOS, eBGP to its PE inside the VRF), two Alpine Linux hosts per DC (h1 in tenant-a, h2 in tenant-b)                                 |                                                                              |
| Resilience | BFD on every core adjacency (300 ms × 3)                                                                                                        | <code>protocols bfd</code>                                                   |

Addresses to keep in mind: PE loopbacks `fd00:a::1-4`, P loopbacks `fd00:a::11-13`; locators `fd00:c:<n>::/48` with the same *n*; tenant-a LANs `172.20.<dc>.0/24`, tenant-b LANs `172.21.<dc>.0/24`; hosts are `.2`, CEs `.1`. Every tenant prefix has an IPv6 twin by rule — `172.X.Y.0 ↔ fd00:X:Y::/64` — so dc3's tenant-a LAN is also `fd00:20:3::/64`, host `::2`.

### 3. The underlay: IS-IS carries the locators

A PE has two core links and therefore two IS-IS adjacencies:

```
vyos@pe1:~$ show isis neighbor
Area VyOS:
System Id      Interface  L  State      Holdtime  SNPA
p1             eth1      2  Up         29        2020.2020.2020
p2             eth2      2  Up         28        2020.2020.2020
```

The locator is a /48 under the block, status Up:

```
vyos@pe1:~$ show segment-routing srv6 locator
Locator:
Name           ID      Prefix              Status
-----
main           1 fd00:c:1::/48      Up
```

IS-IS advertises it as a prefix like any other, plus SRv6 sub-TLVs describing what the node can do with a segment list (how deep an SRH it can process, how many segments it can push). p1 sees all seven SRv6-capable nodes:

```
vyos@p1:~$ show isis segment-routing srv6 node
```

```
Area VyOS:
```

```
IS-IS L1 SRv6-Nodes:
```

```
IS-IS L2 SRv6-Nodes:
```

| System ID      | Algorithm | SRH Max | SL | SRH Max | End Pop | SRH Max | H.encaps | SRH Max | End D |
|----------------|-----------|---------|----|---------|---------|---------|----------|---------|-------|
| 0000.0000.0001 | SPF       | 3       | 3  |         | 2       |         | 5        |         |       |
| 0000.0000.0002 | SPF       | 3       | 3  |         | 2       |         | 5        |         |       |
| 0000.0000.0003 | SPF       | 3       | 3  |         | 2       |         | 5        |         |       |
| 0000.0000.0004 | SPF       | 3       | 3  |         | 2       |         | 5        |         |       |
| 0000.0000.0011 | SPF       | 3       | 3  |         | 2       |         | 5        |         |       |
| 0000.0000.0012 | SPF       | 3       | 3  |         | 2       |         | 5        |         |       |
| 0000.0000.0013 | SPF       | 3       | 3  |         | 2       |         | 5        |         |       |

From pe1's point of view every other locator is an ordinary IS-IS route. Note the metrics:

`fd00:c:11::/48` (p1) and `fd00:c:12::/48` (p2) are one hop away (10), the other PEs two hops (20):

```
vyos@pe1:~$ show ipv6 route isis | grep fd00:c:
```

```
I>* fd00:c:1::/48 [115/0] is directly connected, dum0, seg6local uN, weight 1, 00:37:38
```

```
I>* fd00:c:1:e002::/64 [115/0] is directly connected, eth2, seg6local uA nh6 fe80::5054:ff:fec6:603, e  
th2, weight 1, 00:37:38
```

```
I>* fd00:c:1:e003::/64 [115/0] is directly connected, eth1, seg6local uA nh6 fe80::5054:ff:fec6:503, e  
th1, weight 1, 00:37:38
```

```
I>* fd00:c:2::/48 [115/20] via fe80::5054:ff:fec6:503, eth1, weight 1, 00:37:11
```

```
I>* fd00:c:2::1/128 [115/30] via fe80::5054:ff:fec6:503, eth1, weight 1, 00:37:11
```

```
I>* fd00:c:3::/48 [115/20] via fe80::5054:ff:fec6:603, eth2, weight 1, 00:37:12
```

```
I>* fd00:c:3::1/128 [115/30] via fe80::5054:ff:fec6:603, eth2, weight 1, 00:37:12
```

```
I>* fd00:c:4::/48 [115/20] via fe80::5054:ff:fec6:603, eth2, weight 1, 00:37:12
```

```
I>* fd00:c:4::1/128 [115/30] via fe80::5054:ff:fec6:603, eth2, weight 1, 00:37:12
```

```
I>* fd00:c:11::/48 [115/10] via fe80::5054:ff:fec6:503, eth1, weight 1, 00:37:37
```

```
I>* fd00:c:11::1/128 [115/20] via fe80::5054:ff:fec6:503, eth1, weight 1, 00:37:37
```

```
I>* fd00:c:12::/48 [115/10] via fe80::5054:ff:fec6:603, eth2, weight 1, 00:37:37
```

## Local SIDs: what a node will *do*

IS-IS also installs the node's own SIDs into the Linux kernel as `seg6local` routes. This is the part worth staring at, because it is the entire SRv6 data plane of a P router:

```
vyos@p2:~$ ip -6 route show | grep seg6local
```

```
fd00:c:12:e000::/64 nhid 29 encap seg6local action End.X nh6 fe80::5054:ff:fec6:501 oif eth1 flavors
```

```
next-csid lblen 32 nflen 16 dev eth1 proto isis metric 20 pref medium
```

```
fd00:c:12:e001::/64 nhid 31 encap seg6local action End.X nh6 fe80::5054:ff:fec6:202 oif eth4 flavors
```

```
next-csid lblen 32 nflen 16 dev eth4 proto isis metric 20 pref medium
```

```
fd00:c:12:e002::/64 nhid 33 encap seg6local action End.X nh6 fe80::5054:ff:fec6:102 oif eth3 flavors
```

```
next-csid lblen 32 nflen 16 dev eth3 proto isis metric 20 pref medium
```

```
fd00:c:12:e003::/64 nhid 35 encap seg6local action End.X nh6 fe80::5054:ff:fec6:702 oif eth2 flavors
```

```
next-csid lblen 32 nflen 16 dev eth2 proto isis metric 20 pref medium
```

```
fd00:c:12:e004::/64 nhid 37 encap seg6local action End.X nh6 fe80::5054:ff:fec6:301 oif eth5 flavors
```

```
next-csid lblen 32 nflen 16 dev eth5 proto isis metric 20 pref medium
```

```
fd00:c:12:e005::/64 nhid 39 encap seg6local action End.X nh6 fe80::5054:ff:fec6:401 oif eth6 flavors
```

```
next-csid lblen 32 nflen 16 dev eth6 proto isis metric 20 pref medium
```

```
fd00:c:12::/48 nhid 25 encap seg6local action End flavors next-csid lblen 32 nflen 16 dev dum0 proto
```

```
isis metric 20 pref medium
```

- `fd00:c:12::/48 ... action End flavors next-csid` — the **uN** SID: anything addressed inside my locator that I do not have a more specific entry for is "shift the next uSID into place and forward" (see §6).

- `fd00:c:12:e00x::/64 ... action End.X nh6 ... oif ethN` — one **uA** SID per adjacency: "send it out this link, to this neighbour", regardless of the IGP's opinion.

A P router has nothing else: no VRFs, no tenant routes, no BGP.

```
vyos@p2:~$ ip route show vrf tenant-a; ip vrf show
Error: argument "tenant-a" is wrong: Invalid VRF
```

```
Name          Table
-----
No VRF has been configured
```

A PE has the same two kinds plus one SID per tenant VRF, installed by BGP:

```
vyos@pe1:~$ ip -6 route show | grep seg6local
fd00:c:1:e000:: nhid 45  encap seg6local action End.DT46 vrftable tenant-b dev tenant-b proto bgp metric 20 pref medium
fd00:c:1:e001:: nhid 39  encap seg6local action End.DT46 vrftable tenant-a dev tenant-a proto bgp metric 20 pref medium
fd00:c:1:e002::/64 nhid 42  encap seg6local action End.X nh6 fe80::5054:ff:fec6:603 oif eth2 flavors next-csid lblen 32 nflen 16 dev eth2 proto isis metric 20 pref medium
fd00:c:1:e003::/64 nhid 43  encap seg6local action End.X nh6 fe80::5054:ff:fec6:503 oif eth1 flavors next-csid lblen 32 nflen 16 dev eth1 proto isis metric 20 pref medium
fd00:c:1::/48 nhid 37  encap seg6local action End flavors next-csid lblen 32 nflen 16 dev dum0 proto isis metric 20 pref medium
```

`fd00:c:1:e001:: ... End.DT46 vrftable tenant-a` is the **uDT46** SID: "decapsulate, then route the packet inside — IPv4 or IPv6 — in VRF tenant-a". One SID per VRF serves both families; it is what the remote PEs will put on packets for dc1's tenant-a LANs. The function value (`e001`) is allocated by FRR at run time, which is why the tests and tools read it back rather than assume it.

BFD watches each adjacency so a silent link failure is detected in under a second:

```
vyos@pe1:~$ show bfd peers brief
Session count: 2
SessionId  LocalAddress          PeerAddress          Status
Profile
=====
2026629426 fe80::5054:ff:fec6:101 fe80::5054:ff:fec6:503 up
-
3077310471 fe80::5054:ff:fec6:102 fe80::5054:ff:fec6:603 up
-
```

## 4. The overlay: BGP VPNv4 with SIDs instead of labels

Each PE peers with **both** route reflectors over IPv6 loopbacks. The IPv4-VPN routes travel over IPv6 sessions thanks to the extended-nexthop capability:

```
vyos@pe1:~$ show bgp ipv4 vpn summary
BGP router identifier 10.255.0.1, local AS number 65000 VRF default vrf-id 0
BGP table version 0
RIB entries 15, using 2400 bytes of memory
Peers 2, using 58 KiB of memory
```

| Neighbor                      | V | AS    | MsgRcvd | MsgSent | TblVer | InQ | OutQ | Up/Down  | State/PfxRcd | PfxSnt | Des |
|-------------------------------|---|-------|---------|---------|--------|-----|------|----------|--------------|--------|-----|
| fd00:a::11<br>route reflector | 4 | 65000 | 75      | 49      | 17     | 0   | 0    | 00:37:16 | 46           | 14     | p1  |
| fd00:a::13<br>route reflector | 4 | 65000 | 76      | 51      | 17     | 0   | 0    | 00:37:16 | 46           | 14     | p3  |

Total number of neighbors 2

Inside VRF tenant-a the PE runs plain eBGP with the CE — one session per address family — and the CE advertises its LANs:

```
vyos@pe1:~$ show ip bgp vrf tenant-a summary

IPv4 Unicast Summary:
BGP router identifier 10.255.0.1, local AS number 65000 VRF tenant-a vrf-id 10
BGP table version 50
RIB entries 98, using 15 KiB of memory
Peers 1, using 29 KiB of memory
```

| Neighbor                 | V | AS    | MsgRcvd | MsgSent | TblVer | InQ | OutQ | Up/Down  | State/PfxRcd | PfxSnt | Des |
|--------------------------|---|-------|---------|---------|--------|-----|------|----------|--------------|--------|-----|
| 172.16.1.2<br>(tenant-a) | 4 | 65001 | 47      | 49      | 50     | 0   | 0    | 00:37:35 | 11           | 50     | ce1 |

Total number of neighbors 1

```
vyos@pe1:~$ show bgp vrf tenant-a ipv6 summary

IPv6 Unicast Summary:
BGP router identifier 10.255.0.1, local AS number 65000 VRF tenant-a vrf-id 10
BGP table version 8
RIB entries 15, using 2400 bytes of memory
Peers 1, using 29 KiB of memory
```

| Neighbor                   | V | AS    | MsgRcvd | MsgSent | TblVer | InQ | OutQ | Up/Down  | State/PfxRcd | PfxSnt | Des |
|----------------------------|---|-------|---------|---------|--------|-----|------|----------|--------------|--------|-----|
| fd00:16:1::2<br>(tenant-a, | 4 | 65001 | 46      | 48      | 8      | 0   | 0    | 00:37:42 | 1            | 8      | ce1 |

Total number of neighbors 1

The reflector sees every site of every tenant, distinguished by route distinguisher ( 65000:1xx = tenant-a, 65000:2xx = tenant-b, xx = the PE) and keyed for import by route target:

```
vyos@p1:~$ show bgp ipv4 vpn summary
BGP router identifier 10.255.0.11, local AS number 65000 VRF default vrf-id 0
BGP table version 0
RIB entries 15, using 2400 bytes of memory
Peers 5, using 146 KiB of memory
Peer groups 1, using 72 bytes of memory
```

| Neighbor        | V | AS    | MsgRcvd | MsgSent | TblVer | InQ | OutQ | Up/Down  | State/PfxRcd | PfxSnt | Des |
|-----------------|---|-------|---------|---------|--------|-----|------|----------|--------------|--------|-----|
| fd00:a::1       | 4 | 65000 | 49      | 75      | 14     | 0   | 0    | 00:37:16 | 14           | 60     | pe1 |
| fd00:a::2       | 4 | 65000 | 49      | 75      | 14     | 0   | 0    | 00:37:16 | 14           | 60     | pe2 |
| fd00:a::3       | 4 | 65000 | 49      | 75      | 14     | 0   | 0    | 00:37:16 | 14           | 60     | pe3 |
| fd00:a::4       | 4 | 65000 | 53      | 75      | 14     | 0   | 0    | 00:37:16 | 18           | 60     | pe4 |
| fd00:b:0:121::2 | 4 | 65000 | 230     | 264     | 14     | 0   | 0    | 00:37:44 | 0            | 60     | lg  |

(BGP looking)

Total number of neighbors 5

Now the interesting attribute. Here is dc3's tenant-a LAN as pe1 learned it — twice, once per reflector:

```
vyos@pe1:~$ show bgp ipv4 vpn 172.20.3.0/24
BGP routing table entry for 65000:103:172.20.3.0/24, version 2
not allocated
Paths: (2 available, best #1)
  Not advertised to any peer
  65003
    0.0.0.0 (metric 20) from fd00:a::11 (10.255.0.3)
      Origin IGP, metric 0, localpref 100, valid, internal, multipath, best (Neighbor IP)
      Extended Community: RT:65000:100
      Originator: 10.255.0.3, Cluster list: 10.255.0.11
      Remote labels: 917520
      Remote SID: fd00:c:3::, sid structure=[32 16 16 0 16 48]
      Last update: Tue Sep 29 11:37:16 2026
  65003
    0.0.0.0 (metric 20) from fd00:a::13 (10.255.0.3)
      Origin IGP, metric 0, localpref 100, valid, internal, multipath
      Extended Community: RT:65000:100
      Originator: 10.255.0.3, Cluster list: 10.255.0.13
      Remote labels: 917520
      Remote SID: fd00:c:3::, sid structure=[32 16 16 0 16 48]
      Last update: Tue Sep 29 11:37:16 2026
```

Three things to read off this:

1. Remote SID: fd00:c:3::, sid structure=[32 16 16 0 16 48] — the BGP Prefix-SID attribute. The locator is pe3's, the structure says "32-bit block, 16-bit node, 16-bit function, transposition of 16 bits at offset 48". The function bits were *transposed* into the label field (Remote labels: 917504 = 0xE000 << 4), a standard trick to keep the SID attribute compact; the receiver reassembles fd00:c:3:e001::.
2. RT:65000:100 — the route target; only VRFs importing 65000:100 (tenant-a) get this route. tenant-b never sees it.
3. Originator: 10.255.0.3, Cluster list: 10.255.0.11 / 10.255.0.13 — two copies, one via each reflector, so losing p1 loses nothing (test suite 06 proves it).

The same site's IPv6 LAN arrives as a VPNv6 route — with the **same** SID and the same transposed label, because the SID is allocated per VRF (sid vpn per-vrf export auto), not per address family:

```

vyos@pel:~$ show bgp ipv6 vpn fd00:20:3::/64
BGP routing table entry for 65000:103:fd00:20:3::/64, version 2
not allocated
Paths: (2 available, best #1)
  Not advertised to any peer
  65003
    fd00:a::3 (metric 20) from fd00:a::11 (10.255.0.3)
      Origin IGP, metric 0, localpref 100, valid, internal, multipath, best (Neighbor IP)
      Extended Community: RT:65000:100
      Originator: 10.255.0.3, Cluster list: 10.255.0.11
      Remote labels: 917520
      Remote SID: fd00:c:3::, sid structure=[32 16 16 0 16 48]
      Last update: Tue Sep 29 11:37:16 2026
  65003
    fd00:a::3 (metric 20) from fd00:a::13 (10.255.0.3)
      Origin IGP, metric 0, localpref 100, valid, internal, multipath
      Extended Community: RT:65000:100
      Originator: 10.255.0.3, Cluster list: 10.255.0.13
      Remote labels: 917520
      Remote SID: fd00:c:3::, sid structure=[32 16 16 0 16 48]
      Last update: Tue Sep 29 11:37:16 2026

```

BGP hands the route to zebra, which resolves the SID's locator through IS-IS and installs an **encapsulating** route in the tenant VRF. Compare with the label-based world: there is no label table, the VPN route simply says "wrap in IPv6 to this address":

```
vyos@pel:~$ ip route show vrf tenant-a
default nhid 105 encap seg6 mode encap segs 1 [ fd00:c:4:e001:: ] via inet6 fe80::5054:ff:fec6:603 de
v eth2 proto bgp metric 20
127.0.0.0/8 dev tenant-a proto kernel scope link src 127.0.0.1
172.16.1.0/30 dev eth3 proto kernel scope link src 172.16.1.1
172.16.2.0/30 nhid 97 proto bgp metric 20
    nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:503 dev
eth1 weight 1
    nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:603 dev
eth2 weight 1
172.16.3.0/30 nhid 102 encap seg6 mode encap segs 1 [ fd00:c:3:e001:: ] via inet6 fe80::5054:ff:fec6:
603 dev eth2 proto bgp metric 20
172.16.4.0/30 nhid 105 encap seg6 mode encap segs 1 [ fd00:c:4:e001:: ] via inet6 fe80::5054:ff:fec6:
603 dev eth2 proto bgp metric 20
172.16.5.0/30 nhid 105 encap seg6 mode encap segs 1 [ fd00:c:4:e001:: ] via inet6 fe80::5054:ff:fec6:
603 dev eth2 proto bgp metric 20
172.20.1.0/24 nhid 79 via 172.16.1.2 dev eth3 proto bgp metric 20
172.20.2.0/24 nhid 97 proto bgp metric 20
    nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:503 dev
eth1 weight 1
    nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:603 dev
eth2 weight 1
172.20.3.0/24 nhid 102 encap seg6 mode encap segs 1 [ fd00:c:3:e001:: ] via inet6 fe80::5054:ff:fec6:
603 dev eth2 proto bgp metric 20
172.20.4.0/24 nhid 105 encap seg6 mode encap segs 1 [ fd00:c:4:e001:: ] via inet6 fe80::5054:ff:fec6:
603 dev eth2 proto bgp metric 20
172.24.1.1 nhid 79 via 172.16.1.2 dev eth3 proto bgp metric 20
172.24.1.2 nhid 79 via 172.16.1.2 dev eth3 proto bgp metric 20
172.24.1.3 nhid 79 via 172.16.1.2 dev eth3 proto bgp metric 20
172.24.1.4 nhid 79 via 172.16.1.2 dev eth3 proto bgp metric 20
172.24.1.5 nhid 79 via 172.16.1.2 dev eth3 proto bgp metric 20
172.24.1.6 nhid 79 via 172.16.1.2 dev eth3 proto bgp metric 20
172.24.1.7 nhid 79 via 172.16.1.2 dev eth3 proto bgp metric 20
172.24.1.8 nhid 79 via 172.16.1.2 dev eth3 proto bgp metric 20
172.24.1.9 nhid 79 via 172.16.1.2 dev eth3 proto bgp metric 20
172.24.1.10 nhid 79 via 172.16.1.2 dev eth3 proto bgp metric 20
172.24.2.1 nhid 97 proto bgp metric 20
    nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:503 dev
eth1 weight 1
    nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:603 dev
eth2 weight 1
172.24.2.2 nhid 97 proto bgp metric 20
    nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:503 dev
eth1 weight 1
    nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:603 dev
eth2 weight 1
172.24.2.3 nhid 97 proto bgp metric 20
    nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:503 dev
eth1 weight 1
    nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:603 dev
eth2 weight 1
172.24.2.4 nhid 97 proto bgp metric 20
    nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:503 dev
eth1 weight 1
    nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:603 dev
eth2 weight 1
172.24.2.5 nhid 97 proto bgp metric 20
    nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:503 dev
eth1 weight 1
    nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:603 dev
eth2 weight 1
172.24.2.6 nhid 97 proto bgp metric 20
    nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:503 dev
eth1 weight 1
    nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:603 dev
eth2 weight 1
172.24.2.7 nhid 97 proto bgp metric 20
    nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:503 dev
eth1 weight 1
    nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:603 dev
eth2 weight 1
172.24.2.8 nhid 97 proto bgp metric 20
```



```

        nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:503 dev
eth1 weight 1
        nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:603 dev
eth2 weight 1
172.24.2.9 nhid 97 proto bgp metric 20
        nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:503 dev
eth1 weight 1
        nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:603 dev
eth2 weight 1
172.24.2.10 nhid 97 proto bgp metric 20
        nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:503 dev
eth1 weight 1
        nexthop encap seg6 mode encap segs 1 [ fd00:c:2:e001:: ] via inet6 fe80::5054:ff:fec6:603 dev
eth2 weight 1
172.24.3.1 nhid 102 encap seg6 mode encap segs 1 [ fd00:c:3:e001:: ] via inet6 fe80::5054:ff:fec6:603
dev eth2 proto bgp metric 20
172.24.3.2 nhid 102 encap seg6 mode encap segs 1 [ fd00:c:3:e001:: ] via inet6 fe80::5054:ff:fec6:603
dev eth2 proto bgp metric 20
172.24.3.3 nhid 102 encap seg6 mode encap segs 1 [ fd00:c:3:e001:: ] via inet6 fe80::5054:ff:fec6:603
dev eth2 proto bgp metric 20
172.24.3.4 nhid 102 encap seg6 mode encap segs 1 [ fd00:c:3:e001:: ] via inet6 fe80::5054:ff:fec6:603
dev eth2 proto bgp metric 20
172.24.3.5 nhid 102 encap seg6 mode encap segs 1 [ fd00:c:3:e001:: ] via inet6 fe80::5054:ff:fec6:603
dev eth2 proto bgp metric 20
172.24.3.6 nhid 102 encap seg6 mode encap segs 1 [ fd00:c:3:e001:: ] via inet6 fe80::5054:ff:fec6:603
dev eth2 proto bgp metric 20
172.24.3.7 nhid 102 encap seg6 mode encap segs 1 [ fd00:c:3:e001:: ] via inet6 fe80::5054:ff:fec6:603
dev eth2 proto bgp metric 20
172.24.3.8 nhid 102 encap seg6 mode encap segs 1 [ fd00:c:3:e001:: ] via inet6 fe80::5054:ff:fec6:603
dev eth2 proto bgp metric 20
172.24.3.9 nhid 102 encap seg6 mode encap segs 1 [ fd00:c:3:e001:: ] via inet6 fe80::5054:ff:fec6:603
dev eth2 proto bgp metric 20
172.24.3.10 nhid 102 encap seg6 mode encap segs 1 [ fd00:c:3:e001:: ] via inet6 fe80::5054:ff:fec6:60
3 dev eth2 proto bgp metric 20
172.24.4.1 nhid 105 encap seg6 mode encap segs 1 [ fd00:c:4:e001:: ] via inet6 fe80::5054:ff:fec6:603
dev eth2 proto bgp metric 20
172.24.4.2 nhid 105 encap seg6 mode encap segs 1 [ fd00:c:4:e001:: ] via inet6 fe80::5054:ff:fec6:603
dev eth2 proto bgp metric 20
172.24.4.3 nhid 105 encap seg6 mode encap segs 1 [ fd00:c:4:e001:: ] via inet6 fe80::5054:ff:fec6:603
dev eth2 proto bgp metric 20
172.24.4.4 nhid 105 encap seg6 mode encap segs 1 [ fd00:c:4:e001:: ] via inet6 fe80::5054:ff:fec6:603
dev eth2 proto bgp metric 20
172.24.4.5 nhid 105 encap seg6 mode encap segs 1 [ fd00:c:4:e001:: ] via inet6 fe80::5054:ff:fec6:603
dev eth2 proto bgp metric 20
172.24.4.6 nhid 105 encap seg6 mode encap segs 1 [ fd00:c:4:e001:: ] via inet6 fe80::5054:ff:fec6:603
dev eth2 proto bgp metric 20
172.24.4.7 nhid 105 encap seg6 mode encap segs 1 [ fd00:c:4:e001:: ] via inet6 fe80::5054:ff:fec6:603
dev eth2 proto bgp metric 20
172.24.4.8 nhid 105 encap seg6 mode encap segs 1 [ fd00:c:4:e001:: ] via inet6 fe80::5054:ff:fec6:603
dev eth2 proto bgp metric 20
172.24.4.9 nhid 105 encap seg6 mode encap segs 1 [ fd00:c:4:e001:: ] via inet6 fe80::5054:ff:fec6:603
dev eth2 proto bgp metric 20
172.24.4.10 nhid 105 encap seg6 mode encap segs 1 [ fd00:c:4:e001:: ] via inet6 fe80::5054:ff:fec6:60
3 dev eth2 proto bgp metric 20

```

172.20.3.0/24 ... encap seg6 mode encap segs 1 [ fd00:c:3:e001:: ] via fe80::... dev eth2 — one segment, out eth2 towards p2 (the shortest path to pe3). 172.20.2.0/24 has **two** next hops because pe2 is equidistant via p1 and p2: ECMP for free, from the IGP.

The IPv6 side of the VRF looks the same — the locator leaks (static), then one encapsulating route per remote IPv6 LAN:

```
vyos@pel:~$ ip -6 route show vrf tenant-a
fd00:c:2::/48 nhid 67 proto static metric 20 pref medium
fd00:c:13::/48 nhid 67 proto static metric 20 pref medium
fd00:c::/32 nhid 67 proto static metric 20 pref medium
fd00:16:1::/64 dev eth3 proto kernel metric 256 pref medium
fd00:16:2::/64 nhid 97 proto bgp metric 20 pref medium
fd00:16:3::/64 nhid 102 proto bgp metric 20 pref medium
fd00:16:4::/64 nhid 105 proto bgp metric 20 pref medium
fd00:20:2::/64 nhid 97 proto bgp metric 20 pref medium
fd00:20:3::/64 nhid 102 proto bgp metric 20 pref medium
fd00:20:4::/64 nhid 105 proto bgp metric 20 pref medium
```

The CE does not know any of this happened. It sees the remote LANs as ordinary eBGP routes from its PE:

```
vyos@cel:~$ show ip route vrf tenant-a bgp
Codes: K - kernel route, C - connected, L - local, S - static,
        R - RIP, O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
        T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
        f - OpenFabric, t - Table-Direct,
        > - selected route, * - FIB route, q - queued, r - rejected, b - backup
        t - trapped, o - offload failure
```

IPv4 unicast VRF tenant-a:

```
B>* 0.0.0.0/0 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:23
B 172.16.1.0/30 [20/0] via 172.16.1.1 inactive, weight 1, 00:37:53
B>* 172.16.2.0/30 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.16.3.0/30 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.16.4.0/30 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.16.5.0/30 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.20.2.0/24 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.20.3.0/24 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.20.4.0/24 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.2.1/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.2.2/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.2.3/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.2.4/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.2.5/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.2.6/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.2.7/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.2.8/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.2.9/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.2.10/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.3.1/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.3.2/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.3.3/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.3.4/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.3.5/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.3.6/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.3.7/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.3.8/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.3.9/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.3.10/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.4.1/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.4.2/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.4.3/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.4.4/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.4.5/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.4.6/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.4.7/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.4.8/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.4.9/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
B>* 172.24.4.10/32 [20/0] via 172.16.1.1, eth1, weight 1, 00:37:35
```

## 5. Packet walk: dc1-h1 → dc3-h1

The host has one interface in the tenant LAN and a default route to the CE:

```
lab@dc1-h1:~$ ip -br addr; ip route
lo                UNKNOWN      127.0.0.1/8 ::1/128
eth0              UP           10.3.0.41/24 fe80::5054:ff:fec6:c00/64
eth1              UP           172.20.1.2/24 fd00:20:1::2/64 fe80::5054:ff:fec6:c01/64
default via 172.20.1.1 dev eth1
10.0.0.0/8 via 10.3.0.1 dev eth0
10.3.0.0/24 dev eth0 proto kernel scope link src 10.3.0.41
172.20.1.0/24 dev eth1 proto kernel scope link src 172.20.1.2
```

```
lab@dc1-h1:~$ ping -c 3 172.20.3.2
PING 172.20.3.2 (172.20.3.2): 56 data bytes
64 bytes from 172.20.3.2: seq=0 ttl=42 time=1.970 ms
64 bytes from 172.20.3.2: seq=1 ttl=42 time=1.966 ms
64 bytes from 172.20.3.2: seq=2 ttl=42 time=2.039 ms

--- 172.20.3.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.966/1.991/2.039 ms
```

```
dc1-h1 — ce1 — pe1 —== p2 ==— pe3 — ce3 — dc3-h1
          IPv4      | encap          | decap      IPv4
                    | IPv6 fd00:a::1 → fd00:c:3:e001::
                    | (no SRH: one segment = the destination)
                    |————— plain IPv6 forwarding on p2 —————|
```

Step by step:

1. **ce1** routes `172.20.3.0/24` to **pe1** over the VRF's eBGP session (plain IPv4).
2. **pe1** looks the destination up in VRF tenant-a and hits the `encap seg6` route: it pushes an outer IPv6 header, source = its loopback `fd00:a::1`, destination = **pe3**'s uDT46 SID `fd00:c:3:e001::`. With a single segment the Linux implementation adds an SRH with that one entry (segments-left 0) — functionally the destination address is the whole instruction. Outer lookup: `fd00:c:3::/48` via IS-IS → **eth2** → **p2**.
3. **p2** receives an IPv6 packet for `fd00:c:3:e001::`. It is not in **p2**'s locator, so **p2** does what any IPv6 router does: longest match → `fd00:c:3::/48` → **eth5** → **pe3**. **p2** never looks at the SRH, never knows there is a tenant inside. Captured on **p2**'s link to **pe3**:

```
vyos@p2:~$ sudo tcpdump -ni eth5 -vv 'ip6 and dst net fd00:c:3::/48'
12:15:03.387621 IP6 (hlim 62, next-header Routing (43) payload length: 108) fd00:a::1 > fd00:c:3:e001::: RT6 (len=2, type=4, segleft=0, last-entry=0, flags=0x0, tag=0, [0]fd00:c:3:e001::) IP (tos 0x0, ttl 63, id 40595, offset 0, flags [DF], proto ICMP (1), length 84)
    172.20.1.2 > 172.20.3.2: ICMP echo request, id 103, seq 0, length 64
12:15:03.887745 IP6 (hlim 62, next-header Routing (43) payload length: 108) fd00:a::1 > fd00:c:3:e001::: RT6 (len=2, type=4, segleft=0, last-entry=0, flags=0x0, tag=0, [0]fd00:c:3:e001::) IP (tos 0x0, ttl 63, id 40650, offset 0, flags [DF], proto ICMP (1), length 84)
    172.20.1.2 > 172.20.3.2: ICMP echo request, id 103, seq 1, length 64
12:15:04.387867 IP6 (hlim 62, next-header Routing (43) payload length: 108) fd00:a::1 > fd00:c:3:e001::: RT6 (len=2, type=4, segleft=0, last-entry=0, flags=0x0, tag=0, [0]fd00:c:3:e001::) IP (tos 0x0, ttl 63, id 40929, offset 0, flags [DF], proto ICMP (1), length 84)
    172.20.1.2 > 172.20.3.2: ICMP echo request, id 103, seq 2, length 64
12:15:04.887947 IP6 (hlim 62, next-header Routing (43) payload length: 108) fd00:a::1 > fd00:c:3:e001::: RT6 (len=2, type=4, segleft=0, last-entry=0, flags=0x0, tag=0, [0]fd00:c:3:e001::) IP (tos 0x0, ttl 63, id 41188, offset 0, flags [DF], proto ICMP (1), length 84)
    172.20.1.2 > 172.20.3.2: ICMP echo request, id 103, seq 3, length 64
```

next-header Routing (43), RT6 type=4 is the SRH; segleft=0; the inner packet is the ICMP echo from 172.20.1.2 to 172.20.3.2. Outer hop limit 62 (64 – pe1 – p2); inner TTL 63 (only ce1 decremented it — the core is invisible to the tenant's traceroute, see below).

4. **pe3** owns fd00:c:3:e001:: — its seg6local ... End.DT46 vrf table tenant-a route. The kernel removes the IPv6 header, and routes the IPv4 packet in VRF tenant-a → 172.20.3.0/24 connected via ce3.
5. **ce3** → **dc3-h1**, and the reply does the same in reverse with pe1's uDT46 SID fd00:c:1:e001:: as destination.

The IPv6 tenant does exactly the same, to exactly the same SID — the only difference is what sits inside the outer header:

```
lab@dc1-h1:~$ ping -6 -c 3 fd00:20:3::2
PING fd00:20:3::2 (fd00:20:3::2): 56 data bytes
64 bytes from fd00:20:3::2: seq=0 ttl=61 time=2.717 ms
64 bytes from fd00:20:3::2: seq=1 ttl=61 time=1.913 ms
64 bytes from fd00:20:3::2: seq=2 ttl=61 time=2.001 ms

--- fd00:20:3::2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.913/2.210/2.717 ms
```

```
vyos@p2:~$ sudo tcpdump -ni eth5 -vv 'ip6 and dst net fd00:c:3::/48 and ip6 proto 43'
12:15:10.209606 IP6 (flowlabel 0xd5fc1, hlim 61, next-header Routing (43) payload length: 128) fd00:a::1 > fd00:c:3:e001::: RT6 (len=2, type=4, segleft=0, last-entry=0, flags=0x0, tag=0, [0]fd00:c:3:e001::) IP6 (flowlabel 0xd5fc1, hlim 63, next-header ICMPv6 (58) payload length: 64) fd00:20:1::2 > fd00:20:3::2: [icmp6 sum ok] ICMP6, echo request, id 104, seq 0
12:15:10.709654 IP6 (flowlabel 0xd5fc1, hlim 61, next-header Routing (43) payload length: 128) fd00:a::1 > fd00:c:3:e001::: RT6 (len=2, type=4, segleft=0, last-entry=0, flags=0x0, tag=0, [0]fd00:c:3:e001::) IP6 (flowlabel 0xd5fc1, hlim 63, next-header ICMPv6 (58) payload length: 64) fd00:20:1::2 > fd00:20:3::2: [icmp6 sum ok] ICMP6, echo request, id 104, seq 1
```

IPv6 inside IPv6: fd00:a::1 > fd00:c:3:e001:: carrying fd00:20:1::2 > fd00:20:3::2. That is what End.DT46 buys — dual-stack tenants with one SID, one route per prefix and no second data plane.

The tenant sees exactly one "missing" hop for the whole core (\* at hop 2 is pe1's VRF, which has no address on the core path; the SRv6 hops do not decrement the inner TTL at all):

```
lab@dc1-h1:~$ traceroute -n 172.20.3.2
traceroute to 172.20.3.2 (172.20.3.2), 30 hops max, 46 byte packets
 1  172.20.1.1  0.004 ms
 2  *
 3  172.16.3.2  1.728 ms
 4  172.20.3.2  1.709 ms
```

Isolation between tenants is not a firewall rule — it is the absence of a route. tenant-a's host cannot reach tenant-b's host *at the same site*:

```
lab@dc1-h1:~$ ping -c 2 172.21.1.2
PING 172.21.1.2 (172.21.1.2): 56 data bytes

--- 172.21.1.2 ping statistics ---
2 packets transmitted, 0 packets received, 100% packet loss
```

## 6. uSID: the whole path in one address

The structure `usid-f3216` means: 32-bit **block** `fd00:c`, then 16-bit **node** IDs, then 16-bit **functions**. A locator `fd00:c:3::/48` = block + node 3. And because the node ID is only 16 bits, several can be chained in the remaining 96 bits of one address:

```
fd00:c : 11 : 13 : 3 : e000 ::
-----
block   p1   p3   pe3   uDT46(tenant-b on pe3)
```

Each node whose ID is in the *first* slot after the block owns the packet: its uN SID (End flavors `next-csid`, "NEXT-C-SID") **shifts the address left by 16 bits** (dropping its own ID, padding with zeros) and forwards on the new destination. No SRH entry is consumed; the segment list lives in the address itself.

The lab's steering tool builds exactly that address to force tenant-b's `dc1` → `dc3` traffic over `p1` and `p3` instead of the shortest path through `p2`:

```
$ tools/steer.py add pe1 tenant-b 172.21.3.0/24 p1 p3
pe1: tenant-b 172.21.3.0/24 -> p1 -> p3 -> pe3 segments fd00:c:11:13:3:e000:: (out eth1) [uSID: one compressed segment]
172.21.3.0/24 nhid 142 encap seg6 mode encap segs 1 [ fd00:c:11:13:3:e000:: ] dev eth1 proto static metric 20
```

Watch the destination address change hop by hop. On `p1`'s link *towards* `p3`, `p1` has already consumed its uSID (`:11:`), so the destination is now `fd00:c:13:3:e000::` — while the SRH still shows the original carrier:

```
vyos@p1:~$ sudo tcpdump -ni eth2 -vv 'ip6 and dst net fd00:c::/32'
12:16:19.500122 IP6 (hlim 62, next-header Routing (43) payload length: 108) fd00:a::1 > fd00:c:13:3:e000::: RT6 (len=2, type=4, segleft=0, last-entry=0, flags=0x0, tag=0, [0]fd00:c:11:13:3:e000::) IP (tos 0x0, ttl 63, id 63968, offset 0, flags [DF], proto ICMP (1), length 84)
    172.21.1.2 > 172.21.3.2: ICMP echo request, id 100, seq 0, length 64
12:16:20.001520 IP6 (hlim 62, next-header Routing (43) payload length: 108) fd00:a::1 > fd00:c:13:3:e000::: RT6 (len=2, type=4, segleft=0, last-entry=0, flags=0x0, tag=0, [0]fd00:c:11:13:3:e000::) IP (tos 0x0, ttl 63, id 64088, offset 0, flags [DF], proto ICMP (1), length 84)
    172.21.1.2 > 172.21.3.2: ICMP echo request, id 100, seq 1, length 64
```

On `p3`'s link towards `pe3`, `p3` has shifted its own `:13:` out and the address is down to `pe3`'s uDT46 SID `fd00:c:3:e000::`, exactly what a non-steered packet would carry:

```
vyos@p3:~$ sudo tcpdump -ni eth3 -vv 'ip6 and dst net fd00:c::/32'
12:16:23.624088 IP6 (hlim 63, next-header Routing (43) payload length: 108) fd00:a::3 > fd00:c:4:e000::: RT6 (len=2, type=4, segleft=0, last-entry=0, flags=0x0, tag=0, [0]fd00:c:4:e000::) IP (tos 0x0, ttl 63, id 52453, offset 0, flags [none], proto ICMP (1), length 84)
    172.21.3.2 > 172.21.4.2: ICMP echo reply, id 101, seq 0, length 64
12:16:23.624885 IP6 (hlim 61, next-header Routing (43) payload length: 108) fd00:a::1 > fd00:c:3:e000::: RT6 (len=2, type=4, segleft=0, last-entry=0, flags=0x0, tag=0, [0]fd00:c:11:13:3:e000::) IP (tos 0x0, ttl 63, id 64832, offset 0, flags [DF], proto ICMP (1), length 84)
    172.21.1.2 > 172.21.3.2: ICMP echo request, id 102, seq 0, length 64
```

Hop limit 61 at `p3` = three routers (`pe1`, `p1`, `p3`); the unsteered packet showed 62 after two. The tenant still sees a single opaque hop:

```
lab@dc1-h2:~$ traceroute -n 172.21.3.2
traceroute to 172.21.3.2 (172.21.3.2), 30 hops max, 46 byte packets
 1  172.21.1.1  0.446 ms
 2  *
 3  172.18.3.2  1.751 ms
 4  172.21.3.2  1.804 ms
```

The same path expressed the classic way — three full SIDs in an SRH — also works ( --uncompressed ), and is a good way to see what uSID saves: a three-entry SRH is 56 bytes and every P router on the path has to process it; the uSID packet carries the same one-entry SRH as unsteered traffic (24 bytes, and none at all with reduced encapsulation) and the P routers only rewrite the destination address:

```
$ tools/steer.py add pe1 tenant-b 172.21.3.0/24 p1 p3 --uncompressed
172.21.3.0/24 nhid 168 encap seg6 mode encap segs 3 [ fd00:c:11:: fd00:c:13:: fd00:c:3:e000:: ] dev e
th1 proto static metric 20
```

Removing the policy puts the prefix back on the BGP-learned single-segment route:

```
$ tools/steer.py del pe1 tenant-b 172.21.3.0/24
pe1: tenant-b 172.21.3.0/24 back on the IGP shortest path
172.21.3.0/24 nhid 113 encap seg6 mode encap segs 1 [ fd00:c:3:e000:: ] via inet6 fe80::5054:ff:fec6:
603 dev eth2 proto bgp metric 20
```

## 7. What breaks, and how the lab proves it does not

- **Reflector failure** (suite 06): shutting every client session on p1 leaves every VRF route, every SRv6 route and every in-tenant ping intact via p3 — the two `Cluster List` copies in §4 are the reason.
- **Link failure** (suite 08): a silent cut of p2–pe3 (firewall drop of IS-IS + BFD, not an interface shutdown) is detected by BFD in ~1 s; pe3 moves every tenant route to p3, and a 0.2 s ping stream across cut and repair loses  $\leq 10$  packets (measured: 4). Nothing in BGP changes: the VPN routes still point at the same SIDs, only the IGP path to the locator moved.
- **Throughput** (suite 10): ~140 Mbit/s TCP host to host on a 1-vCPU software data plane over UDP-tunnelled links; steered uSID and uncompressed paths within a few percent of each other.
- **Tenant isolation** (suite 05):  $2 \times 4 \times 3$  in-tenant pings succeed, every cross-tenant pair fails, P routers hold no tenant state.

## 8. Getting out: one firewall for every tenant

A tenant that can only reach its own sites is half a service. The lab adds an internet breakout the way a provider would *not* do it in the textbook and the way it actually works: a small VyOS firewall, `fw-inet`, is a **CE of both tenants on pe4** — one attachment circuit per tenant, each in that tenant's VRF on the firewall too (VRF-lite) — and it announces **one prefix into each: the default route**. Its third port is on the host's libvirt NAT network (DHCP), in its default VRF, where everything



from 172.16.0.0/12 is masqueraded. The default route then travels like any other tenant route: a VPNv4 route under pe4's RD with pe4's End.DT46 SID for that tenant, so on pe1

```
$ ip route show vrf tenant-a default
default nhid 105 encap seg6 mode encap segs 1 [ fd00:c4:e001:: ] via inet6 fe80::5054:ff:fec6:603 de
v eth2 proto bgp metric 20
```

On the firewall the three VRFs are glued together with BGP `import vrf` (no SRv6 here, so plain kernel routes): DHCP's default route (a static in FRR) goes into each tenant VRF, the tenants' routes come into the default VRF for the way back.

```
$ show ip route vrf all | match '0.0.0.0/VRF'
IPv4 unicast VRF default:
S>* 0.0.0.0/0 [210/0] via 192.168.122.1, eth3, weight 1, 00:41:10
IPv4 unicast VRF tenant-a:
B>* 0.0.0.0/0 [210/0] via 192.168.122.1, eth3 (vrf default), weight 1, 00:40:53
IPv4 unicast VRF tenant-b:
B>* 0.0.0.0/0 [210/0] via 192.168.122.1, eth3 (vrf default), weight 1, 00:40:53
```

From a host the path is CE → PE → SRv6 → pe4 → firewall → NAT → the host's uplink:

```
$ traceroute -n -w 1 -q 1 1.1.1.1
traceroute to 1.1.1.1 (1.1.1.1), 6 hops max, 46 byte packets
 1 172.20.1.1 0.377 ms
 2 *
 3 *
 4 *
 5 192.168.50.1 2.224 ms
 6 142.254.153.129 13.709 ms
```

The tenants still never meet. The firewall's forward policy is *established*, then *tenant VRF* → *uplink* per tenant, then drop and log — and anything for 172.16.0.0/12 is dropped first, because a tenant VRF only ever sends the **other** tenant's addresses here (its own are routed inside the VPN). A tenant-a host pinging a tenant-b host shows up in the firewall log, not on the other host:

```
$ show log firewall | match FWD-filter-8
Sep 29 12:17:54 kernel: [ipv4-FWD-filter-8-D]IN=tenant-a OUT=eth3 MAC=52:54:00:c6:14:01:52:54:00:c6:0
4:05:08:00 SRC=172.20.1.2 DST=172.21.2.2 LEN=84 TOS=0x00 PREC=0x00 TTL=61 ID=55847 DF PROTO=ICMP TYPE=
8 CODE=0 ID=109 SEQ=0
Sep 29 12:17:55 kernel: [ipv4-FWD-filter-8-D]IN=tenant-a OUT=eth3 MAC=52:54:00:c6:14:01:52:54:00:c6:0
4:05:08:00 SRC=172.20.1.2 DST=172.21.2.2 LEN=84 TOS=0x00 PREC=0x00 TTL=61 ID=56616 DF PROTO=ICMP TYPE=
8 CODE=0 ID=109 SEQ=1
```

(An "internet VRF" with route-target import/export was tried first and dropped: FRR installs a VPN route leaked locally between two VRFs with the exporting VRF's SRv6 encapsulation, which the kernel cannot forward through an IPv4 next hop, and the internet VRF's routes to every tenant made it a transit path between tenants that no rule on the PE can close — SRv6 re-encapsulation skips the IPv4 forward hook. With the firewall as a plain CE of each tenant, neither problem exists.)

## 9. Operating it like a network, not a demo

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The point of the lab is not the seven routers, it is everything around them:

- **Source of truth** — Nautobot models sites, devices, links, prefixes, VRFs with RTs and RDs, BGP peerings and the SRv6 locators; `nautobot/render.py` renders the exact configuration each node runs, and a test checks the rendering against `lab.conf` and against the routers.
- **Provisioning** — a portal (FastAPI) adds a tenant or a site as a pipeline: allocate, model in Nautobot, render, bring up VMs, push configs, run the tests, back the configurations up to Gitea — with each step's log on the page.
- **Verification** — 81 Robot Framework cases across 14 suites, run after every change; results (with every node's configuration and routing tables) are committed to the repository.
- **Monitoring** — node-exporter and frr-exporter on every router, node-exporter on every host, and the portal's own `/metrics` for what exporters cannot see (tenant health, IS-IS/BFD adjacency counts), scraped by Prometheus into VictoriaMetrics with Grafana dashboards and alert rules:

```
$ curl -s http://10.3.0.11:9342/metrics | grep -E '^frr_(bgp_peer_state|bfd_peer_state)'
frr_bfd_peer_state{local="fe80::5054:ff:fec6:101",peer="fe80::5054:ff:fec6:503"} 1
frr_bfd_peer_state{local="fe80::5054:ff:fec6:102",peer="fe80::5054:ff:fec6:603"} 1
frr_bgp_peer_state{afi="ipv4",local_as="65000",peer="172.16.1.2",peer_as="65001",safi="unicast",vrf="tenant-a"} 1
frr_bgp_peer_state{afi="ipv4",local_as="65000",peer="172.18.1.2",peer_as="65001",safi="unicast",vrf="tenant-b"} 1
frr_bgp_peer_state{afi="ipv4",local_as="65000",peer="fd00:a::11",peer_as="65000",safi="vpn",vrf="default"} 1
frr_bgp_peer_state{afi="ipv4",local_as="65000",peer="fd00:a::13",peer_as="65000",safi="vpn",vrf="default"} 1
```

```
$ curl -s http://127.0.0.1:8091/metrics | grep -E '^lab_(tenant_health|isis_adjacencies_up|tenant_site_bgp_up)'
lab_tenant_site_bgp_up{lab="srv6-core",tenant="tenant-a",dc="dc1",pe="pe1",ce="ce1",external=""} 1
lab_tenant_site_bgp_up{lab="srv6-core",tenant="tenant-b",dc="dc1",pe="pe1",ce="ce1",external=""} 1
lab_tenant_health{lab="srv6-core",tenant="tenant-a",sites="4"} 2
lab_tenant_health{lab="srv6-core",tenant="tenant-b",sites="4"} 2
lab_isis_adjacencies_up{lab="srv6-core",node="pe1",role="pe"} 2
lab_isis_adjacencies_up{lab="srv6-core",node="pe2",role="pe"} 2
lab_isis_adjacencies_up{lab="srv6-core",node="pe3",role="pe"} 2
lab_isis_adjacencies_up{lab="srv6-core",node="pe4",role="pe"} 2
```

## 10. Three things worth remembering

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1. **SRv6 is just IPv6 routing plus local instructions.** The P routers in this lab run IS-IS and forward IPv6; every SRv6 behaviour is a `seg6local` route on the node that owns the address. If you can read `ip -6 route`, you can debug the core.
  2. **The VPN is the same BGP VPN you already know**, with a SID where the label used to be. RD, RT, reflectors, import and export policy all carry over unchanged; only the transport moved from labels to addresses — and with End.DT46, IPv4 and IPv6 tenants share one SID and one route table.
  3. **uSID makes traffic engineering cheap.** A four-hop explicit path costs zero extra bytes on the wire and no state in the core — the path is the destination address, rewritten as it goes.
-



*Everything here is reproducible: <https://github.com/dcantor/srv6-core> — `./lab.sh up && ./lab.sh bootstrap && ./lab.sh test`. Configuration lines are in the *README*; the renderer is `tools/render.py`. Captures: `docs/walkthrough_capture.py`; this document: `docs/build_walkthrough.py`.*